

04/24/2024
Purchase Order: 26627161 Sales Order: 1001000647
Heat Number 24001762 Vendor Name Big River Steel - Osceola AR US Mfg. Date 03/14/2024
Item Number 11000102 Description 5L- X52 4-1/2X.337 (15.00#)BR,DRL,PEB

PAGE: 10

CHEMICAL ANALYSIS

	C	Mn	P	S	Ni	Cr	Cu	Mo	Si	Ti	V	CE
Ladle	0.0600	0.5200	0.0080	0.0040	0.0600	0.1000	0.1600	0.0140	0.1400	0.0000	0.0020	0.1108
	Al:	0.0290	Sn:	0.0056	N:	0.0064	B:	0.0010	Ca:	0.0021	Nb:	0.0120
Chem	0.0742	0.5373	0.0078	0.0032	0.0663	0.0883	0.1369	0.0126	0.1435	0.0012	0.0035	0.1196
	Al:	0.0330	Sn:	0.0052	N:	0.0139	B:	0.0000	Ca:	0.0024	Nb:	0.0144
Chem	0.0630	0.5436	0.0088	0.0033	0.0650	0.0921	0.1356	0.0124	0.1425	0.0007	0.0039	0.1090
	Al:	0.0320	Sn:	0.0053	N:	0.0165	B:	0.0001	Ca:	0.0019	Nb:	0.0152

Acceptance Criteria: CEPCM = 0.2500

MECHANICAL PROPERTIES

YS (ksi): 68.80 TS (ksi): 73.50 EL (%) : 28.00 HRB: 90.00 Y/T(%): 0.94

CHARPY TEST RESULTS BODY

	Specimen(S)	Energy (ft/lbs.)	%Shear
ORIENTATION: TRANSVERSE	1	76.00	100.00
Spec. Size (in) : .394"X.197"X2.165"	2	74.00	100.00
Test Temp,(of) -20	3	71.00	100.00
Location 3T90	Average	73.67	100.00

Acceptance Criteria Min. FT/lbs.: Min. Ft/Lbs 10

COMMENTS

MINIMUM HYDROSTATIC TEST PRESSURE (PSI) : 4,700 PER 5 SECONDS

MANUFACTURED TO THE 46TH EDITION OF API 5L SPECIFICATION
MANUFACTURED IN THE USA-HIGH FREQUENCY WELDED PRODUCT
WELD SEAM ANNEALING AT 1600 DEGREES F.MINIMUM
ULTRASONIC TESTED TO N-10 NOTCH 1/8 DRILLED HOLE
FLATTENING TEST PASSED
1 INCH L90 STRIP SPECIMEN
THIS MATERIAL MEETS :ASTM A53B-22,ASME SA53B-21 LATEST EDITION
API 5L PSL 2 GRADE X52M
MEETS HARDNESS REQUIREMENTS OF NACE MRO 175/ISO 15156-2, CLAUSE A.2.1.1
MEETS ISO 10474 3.1.B/EN 10204: 3.1 / STANDARDS ARE LATEST REVISION
MELTED AND MANUFACTURED IN THE U.S.A.
STEEL SUPPLIER: Big River Steel - Osceola AR US
COIL MANUFACTURER: Big River Steel - Osceola AR US

The above analyses and tensile properties are correct to the best of my knowledge and belief.

Quality Control Dept